ActDelta: Decision Relevance Is Learnable Offline but Does Not Yet Beat Periodic Sampling Online

A Measurement Study of Predictive Uplink Triggering for Edge-Offloaded Robot Policies

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Abstract

Edge-offloaded robot policies incur recurring camera-uplink cost. We test *decision relevance*—the action change caused by replacing a true observation with its prediction—as an alternative to reconstruction fidelity. Across four LIBERO suites, four fidelity metrics correlate weakly with action divergence (pooled Spearman −0.117 to −0.028), while a frozen latent head reaches Spearman 0.684 and AUROC 0.833. This offline gain does not transfer online. Against rate-matched periodic sampling, the learned trigger changes success by +3.13 pp at a 25% budget and −3.75 pp at 50%, with both intervals including zero; a true-divergence oracle also fails, and Bernoulli sampling is not separated from the learned trigger across the sweep. Threshold triggering instead creates a heavier observation-age tail, and outcomes track age more strongly than instantaneous divergence. Decision relevance is learnable but insufficient for closed-loop scheduling.

CCS Concepts

• **Networks** → **Network measurement**; • **Computer systems organization** → *Robotics.*

Keywords

Predictive transmission, event-triggered sampling, world models, vision-language-action models, edge offloading, task-oriented communication, measurement, negative results

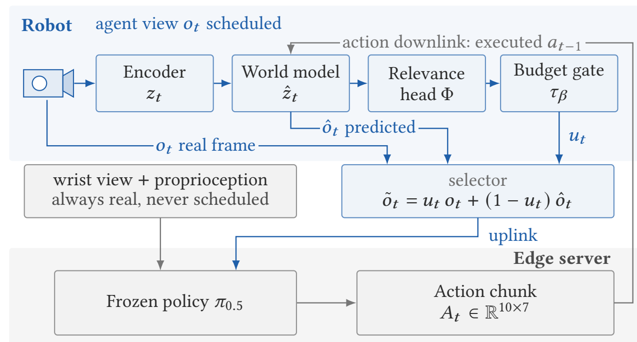

Figure 1: ActDelta schedules the agent view while wrist view and proprioception remain current in closed loop.

1 Introduction

Generalist robot policies remain difficult to host on constrained robots [4, 21, 38]. Edge deployments therefore upload observations and return future-action chunks while the previous chunk executes [8, 12, 19, 24, 37]. Compression reduces bytes but not the question of which observations merit a shared, variable uplink (Figure 1).

Event-triggered communication sends when receiver-side estimation error grows [18, 30, 35, 40]. A robot policy instead consumes observations to act: large visual error may preserve an action, while a small change may alter contact. We therefore learn offline the action-chunk divergence between true and predicted observations, estimate it from local world-model latents, and transmit when it is high.

The experiments reverse the expected success story. First, on 1,400 frozen test pairs covering four LIBERO suites and seven horizons, pixel RMS, SSIM error, LPIPS, and world-model latent distance correlate weakly with action divergence and disagree with its median-threshold decision on 48.9–53.9% of frames. Second, decision relevance itself is learnable: a head that sees frozen latent features but no suite, task, or horizon identifier reaches test Spearman 0.684 (95% CI [0.637, 0.726]), AUROC 0.833, and AUPRC 0.833. Third, this offline skill does not establish a closed-loop advantage. At matched target budgets, the learned trigger differs from periodic sampling by +3.13 pp (CI [−1.25, +7.50]) at 25% and −3.75 pp (CI [−8.13, 0]) at 50%, and neither an oracle trigger that observes true divergence nor a full budget sweep moves the learned arm outside the control's confidence band.

Two controls diagnose the gap. Content-blind Bernoulli sampling tracks the learned arm within resolution while both beat a fidelity trigger, and threshold triggering has a heavier age tail than periodic sampling at equal mean rate. The loop rewards refresh spacing over frame identity.

We contribute a multi-suite fidelity measurement, a frozen-latent relevance predictor, a matched-rate closed-loop test with an oracle, and age-tail and outcome diagnostics. Together they show that per-frame relevance ranking and transmission scheduling are different problems.

2 Background and Motivation

Edge-offloaded robot policies. A chunked vision-language-action policy maps camera observations, proprioception, and a language instruction to a sequence of future actions [4, 11, 51]. In an asynchronous deployment, the robot uploads an observation, receives an action chunk, and overlaps execution with the next request. The uplink carries high-dimensional images, whereas the downlink carries compact action vectors. Remote-policy systems optimize compression, pipelining, caching, and edge collaboration [8, 12, 19, 24, 37], but rarely decide whether a frame is needed. Because cellular capacity varies [33, 34, 39], we study visual-budget allocation rather than end-to-end throughput.

Predictive transmission. Send-on-delta and event-triggered control suppress updates while a shared estimate remains sufficiently accurate [2, 3, 17, 30, 31, 41]. Recent semantic-communication systems extend this idea with learned predictors or generative world models [18, 44, 47, 48]. Although their predictors differ, the triggering rule remains fidelity-centered: transmit when observation error grows. This transfers an estimator objective to a policy receiver without asking whether observation error predicts action change.

Compute asymmetry and the proposed correction. The frozen policy has billions of parameters while the predictor has 13.5M [4, 21, 38]. ActDelta distills the expensive true-versus-predicted action comparison into a latent head; the test is whether its scores beat periodic updates at fixed budget.

Baselines and accounting. Always-send and fixed horizons define the frontier. Fidelity, learned, periodic, Bernoulli, and oracle arms isolate trigger statistic, deployable relevance, rate, spacing, and estimation error, respectively (Figure 6). All act on time and can compose with spatial codecs.

3 Problem Formulation and Claims

The robot produces an agent-view observation o_t at control step t. The edge server either receives that observation or advances a shared predictor to obtain $\hat{o}_t$. A binary scheduling decision $u_t \in \{0,1\}$ selects the server input,

$$\tilde{o}_t = u_t o_t + (1 - u_t)\,\hat{o}_t, \tag{1}$$

where this notation denotes selection rather than pixel interpolation. The frozen policy maps $\tilde{o}_t$, the real wrist view, proprioception, and instruction to an action chunk. The executed action changes the environment, so the decision affects the current action and the future observation distribution.

For an episode of length T, a scheduler g produces decisions $u_{1:T}$ from information locally available at the robot. Its realized agent-view send fraction is $R(g) = \frac{1}{T}\sum_{t=1}^{T} u_t$. Given a target budget b, the systems objective is not to minimize reconstruction error but to maximize task success S subject

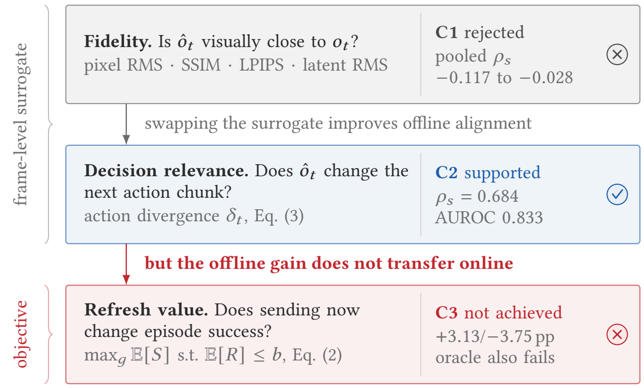

Figure 2: The three questions the paper separates. Fidelity does not rank decision relevance (C1); decision relevance is learnable offline (C2); neither buys refresh value in closed loop (C3).

to the realized communication constraint,

$$\max_{g}\ \mathbb{E}[S(g)] \quad \text{s.t.} \quad \mathbb{E}[R(g)] \le b. \tag{2}$$

This makes periodic sampling indispensable. A relevance trigger is useful only if it improves success at the same realized R, or reduces R at matched success. Comparing methods at different send fractions does not answer Equation (2).

3.1 Four Candidate Scheduling Signals

A **fidelity trigger** thresholds pixel RMS, SSIM error, LPIPS, or latent RMS. A **policy-aware trigger** thresholds action divergence δ_t (Equation (3)); the deployable arm uses $\hat{\delta}_t$, while the oracle observes δ_t. **Periodic** uses time since the last send, whereas independent **Bernoulli** sampling $u_t \sim \mathrm{Bern}(b)$ removes regular spacing.

Fidelity asks whether prediction is visually accurate. Action divergence asks whether replacing prediction with truth changes the next action chunk. Equation (2) asks whether sending now changes eventual success. Figure 2 places the three questions on one axis. The first two are frame-level surrogate objectives; the third is the actual systems objective. The paper's negative result arises because replacing the first surrogate with the second improves offline alignment without closing the remaining gap to the third.

3.2 Layered Claims

The argument is intentionally layered so that each claim is evaluated independently. C1 is a *measurement* claim: fidelity and decision relevance are decoupled in embodied observation streams. It is falsified if fidelity metrics consistently rank action divergence across held-out tasks and horizons. C2 is a *representation* claim: decision relevance can be predicted from the robot-side world-model state without online policy

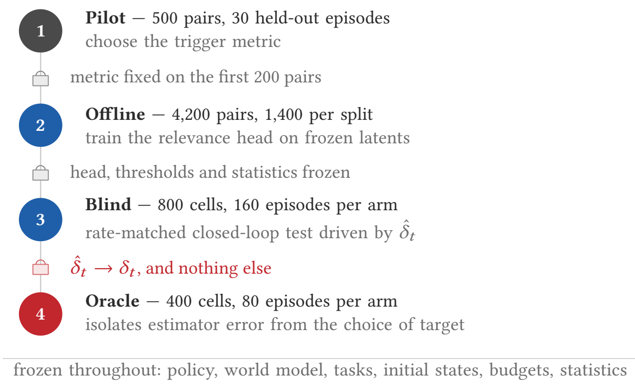

Figure 3: Four frozen stages isolate metric choice, relevance learning, blind testing, and the oracle target.

inference. It is falsified if the head fails on episode-disjoint data or collapses in any major suite or horizon. C3 is the *systems* claim inherited from the positive ActDelta design: allocating transmissions by decision relevance improves the success–communication trade-off over periodic sampling. It requires a frozen, rate-matched closed-loop win and is not rescued by offline correlation.

The data support C1 and C2; C3 fails under the evaluated protocol. This outcome does not collapse the paper into an implementation failure. It identifies a specific substitution error: the original design correctly moves from pixels to decisions, but still uses immediate decision displacement as a proxy for episode-level refresh value. The oracle experiment makes this distinction observable and turns the failed positive claim into the paper's principal finding.

4 Measurement Setup and Protocol

Our protocol separates three questions: whether fidelity tracks action change, whether action change is predictable offline, and whether predicted relevance improves closed-loop success. Figure 3 summarizes this separation.

4.1 Environment and Observation Model

All experiments use LIBERO [26]: Spatial, Object, Goal, and LIBERO-10, with ten tasks per suite. We run one frozen official $\pi_{0.5}$ LIBERO checkpoint through OpenPI at a pinned revision. Paired policy evaluations use common diffusion noise, so their difference isolates the observation substitution rather than independent policy sampling. In a pilot control, repeated inference on an identical observation and identical noise produced maximum RMS difference zero; independent noise produced median 0.0303, compared with 0.5209 for true-versus-predicted observations under common noise.

Table 1: The frozen trajectory base spans ten tasks per suite and five fixed initial states per task; failed episodes are retained and splits do not overlap.

Suite	Success	Transitions	Size
LIBERO-Spatial	49/50 (98%)	5,375	627 MB
LIBERO-Object	48/50 (96%)	7,294	1.1 GB
LIBERO-Goal	48/50 (96%)	5,937	676 MB
LIBERO-10 (Long)	47/50 (94%)	13,289	1.5 GB
Total	192/200 (96%)	31,895	3.9 GB
Splits, assigned by episode index within each suite–task cell			
Train (ep. 0–2)	120 ep.	19,022	
Val (ep. 3)	40 ep.	6,394	
Test (ep. 4)	40 ep.	6,479	

The action-conditioned world model has 13,518,723 parameters, two camera views, 96 tokens per step, width 384, six latent transformer blocks, and 7-D action conditioning. In all closed-loop runs, real 8-D proprioception and the real wrist-camera view remain available. Only the agent view is predicted on steps without transmission. The claim is about agent-view suppression with a real wrist fallback, not full visual replacement; Section 11.3 tests how much it matters.

The predictor starts from the last synchronized observation and advances with the committed action sequence. A transmitted frame replaces the imagined agent view and resets the rollout state; a suppressed step advances the latent prediction. Figure 5 shows one such interval step by step. The evaluated scope is explicit: the server owns policy inference, the robot-side path owns encoding, latent dynamics, and triggering, and the deployable arm makes no policy call.

4.2 Trajectory Base and Splits

The trajectory base contains 200 episodes: four suites, ten tasks per suite, and five fixed initial states per task. The policy succeeds in 192 episodes; failures are retained. Within each suite–task cell, episodes 0–2 form training, episode 3 validation, and episode 4 test, yielding 120/40/40 episodes and 19,022/6,394/6,479 transitions. Episode keys do not overlap. Table 1 is the only dataset table in the main text.

4.3 Action Divergence and Trigger Metrics

Let $A_t, \hat{A}_t \in \mathbb{R}^{K\times d}$ be the action chunks emitted by the pinned policy pipeline for the true and predicted observations under shared noise. In the LIBERO configuration, $K = 10$ and $d = 7$. We define

$$\delta_t = \sqrt{\frac{1}{Kd}\sum_{k=1}^{K}\sum_{j=1}^{d}\left(A_{t,kj} - \hat{A}_{t,kj}\right)^2}. \quad (3)$$

The offline learning pipeline subsequently standardizes divergence per task. This is a reference measure of immediate observation-induced action change, not ground-truth episode utility. Fidelity disagreement thresholds each candidate distance and divergence at validation medians and reports the fraction of test pairs assigned different high/low labels. Communication is measured by agent-view send fraction, by raw-equivalent logical payload, and, in Section 12, by encoded bytes under a fixed codec.

4.4 Freezing and Statistics

Model checkpoints, thresholds, and operating points are selected on validation. The blind test freezes trigger thresholds, code hash, and statistics before execution; earlier exploratory runs remain separate. We report pooled and per-episode correlations. Offline intervals use episode-cluster bootstrap with suite stratification where applicable; closed-loop intervals cluster by task. Because initial states are held fixed across arms, closed-loop comparisons are *paired*, and we report McNemar intervals over the discordant episodes. Target and realized send rates are both reported. An interval covering zero is treated as a failure to establish superiority, not as evidence of equivalence; Section 10.5 quantifies what the design can and cannot resolve.

Experiment records contain repository revisions and SHA-256 digests. Validators check shapes, action–observation alignment, metadata, episode keys, and recomputed metrics. The formal offline study contains 4,200 pairs in total: 1,400 per split, formed by four suites × ten tasks × seven horizons × five pairs. The closed-loop runner fixes initial states across arms and records success, send decisions, observation age, and payload.

Reproducibility of the figures. Every figure is generated from a single tagged data module. No number is typed into a plotting script or into the prose twice: quantities that appear in both are injected into the text at build time. A build-time self-check asserts each cross-panel identity—that success values lie on the grid their episode count can emit, that both binnings in Figure 15 partition the same episodes, and that age distributions integrate to the send fraction.

5 Selecting the Shared Predictor

The predictor is measurement infrastructure rather than an architectural contribution, so we report only the choices that affect the conclusions. An initial twenty-episode gate exposed an action-blind solution: shuffling the action sequence increased latent MSE by only 3.85%. Freezing the encoder and decoder and training a horizon-weighted rollout objective increased action sensitivity to 11.0% and improved five-step error by 46.3%, establishing that the pipeline could learn

Table 2: Only removing the image objective improves rollout at the horizons the trigger uses.

Variant	Outcome	Decision
v1: joint finetune (recon. + image + rollout)	1-step −9.7%; $h5$ +33%; test recon. L_1 +14.0%; action dependence weakened	reject
v2: frozen autoencoder, joint objective retained	test $h5$ 0.0164 → 0.0327 (+99.3%); shuffled/true 1.0875 → 1.0250	reject
v3: frozen AE, rollout objective on random windows	test $h1$ −4.8%, $h5$ −13.2%, $h8$ −17.3%, $h12$ −22.1%; 18,542 windows	**adopt**

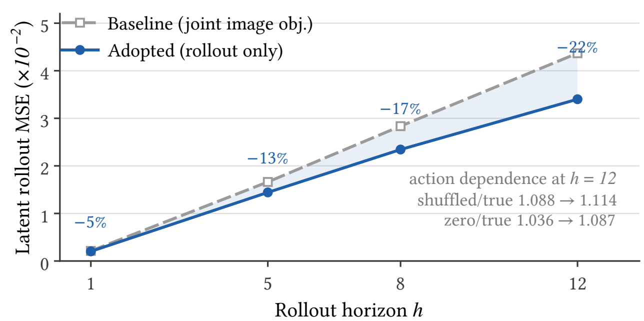

Figure 4: Dropping the image objective increasingly improves rollout error with horizon; one-step reconstruction selects the wrong predictor.

action-conditioned dynamics. Because this gate uses twenty episodes, it establishes feasibility rather than generalization.

Full four-suite selection compared three variants (Table 2). Joint image and rollout objectives improved one-step reconstruction while worsening five-step rollout by 33–99% and weakening action dependence. We therefore adopt the variant trained only on 18,542 randomized latent-rollout windows. Its improvement grows from 4.8% at $h = 1$ to 22.1% at $h = 12$ (Figure 4), although five-step gains remain heterogeneous across suites (4.3–33.4%). At the longest horizon, the shuffled-to-true ratio rises from 1.088 to 1.114 and the zero-to-true ratio from 1.036 to 1.087: the adopted model improves rollout while becoming more dependent on the executed actions, which is the property the counterfactual trigger requires. The result warns that image appearance and one-step accuracy can select the wrong predictor for multi-step action-conditioned use.

6 ActDelta Trigger Design

ActDelta uses a fixed system architecture: the robot and server share an action-conditioned world model; the robot estimates whether the real observation would change the remote policy's action; a controller maps that score to the available transmission budget; and periodic resynchronization prevents indefinite open-loop prediction. Our evaluation separates the components current experiments support from the full loss-recovery protocol deployment would require.

6.1 Latent Decision Head

At a synchronized step, the robot encodes the real observation into z_t. Between synchronizations, the world model produces $\hat{z}_t$ from the last shared latent and the actions already committed for execution. A learned head consumes both representations and their signed difference,

$$\hat{\delta}_t = \Phi(z_t,\ \hat{z}_t,\ z_t - \hat{z}_t), \tag{4}$$

and approximates the offline counterfactual in Equation (3). Retaining both latents lets the head learn context-dependent sensitivity that a norm $\|z_t - \hat{z}_t\|$ discards. The head never receives a task label, suite label, prediction horizon, or policy output. Training therefore distills a policy-in-the-loop measurement into a robot-side statistic without making online policy inference part of the trigger.

6.2 Budgeted Transmission Rule

The deployable rule transmits when the relevance score exceeds a threshold selected for a target budget,

$$u_t = \mathbf{1}[\hat{\delta}_t > \tau_b]\ \vee\ \mathbf{1}[k_t > K_{\max}], \tag{5}$$

where u_t is the agent-view send decision, b is the target send fraction, and k_t counts steps since the last real agent view. The first clause allocates transmissions by predicted relevance; the second imposes the maximum-silence safeguard used in the experimental runner, so at most $K_{\max}$ consecutive steps may be suppressed and the largest reachable agent-view age is exactly $K_{\max}$. Thresholds are chosen on validation and frozen before the blind test. The causal budget controller then tracks realized sends without consulting future frames. Its measured errors—0.13 and 0.08 percentage points at the two blind-test budgets—show that the comparison isolates scheduling from transmission quantity.

Periodic sampling replaces the first clause with a fixed schedule while retaining the same observation model and runner; Bernoulli sampling replaces it with an unbiased coin at rate b; the oracle replaces $\hat{\delta}_t$ by δ_t and is deliberately undeployable. These one-variable substitutions are essential: all arms otherwise share the policy, predictor, tasks, initial states, and budget accounting. Figure 6 states which comparison isolates which variable.

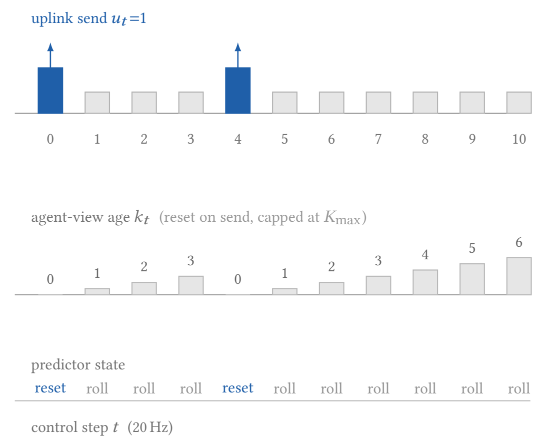

Figure 5: A send resets prediction and age; suppression advances both. Periodic and Bernoulli controls differ only in placement at equal mean rate.

	sees scene content	regular spacing	perfect δ_t
Periodic (control)	no	yes	–
Bernoulli	no	no	–
LPIPS threshold	yes	no	–
ActDelta learned	yes	no	no
Oracle δ_t	yes	no	yes

Each pair differing in exactly one column is a diagnostic: **learned – oracle** isolates estimation error; **oracle – periodic** isolates the choice of target; **Bernoulli – periodic** isolates the value of bounded silence; **LPIPS – learned** isolates the trigger statistic.

Figure 6: The control matrix isolates one substitution per stage.

6.3 Synchronization Boundary

The current runner forces refresh after a bounded silence interval and resets the predictor when a frame is transmitted. It does not implement a complete sequence-number, acknowledgement, retransmission, and action-hash protocol. We therefore make no packet-loss robustness claim. In a real network, silence generated by Equation (5) is observationally identical to an uplink loss unless the protocol carries independent liveness state [36]. Sequence numbers, lightweight heartbeats, server-requested resynchronization, and a safe always-send fallback remain necessary engineering extensions. This boundary does not affect the no-loss matched-budget comparison, but it prevents the current prototype from being presented as a complete transport.

7 Fidelity Does Not Rank Relevance

The formal protocol has four suites, ten tasks, seven horizons $\{1, 2, 3, 4, 5, 8, 12\}$, and five pairs per cell: 1,400 pairs in each episode-disjoint train, validation, and test split. Pixel RMS, SSIM error, LPIPS, and world-model latent RMS are evaluated against Equation (3); thresholds come from validation.

Before the formal study, a 500-pair pilot over 30 held-out episodes checks label quality and statistical power. Its divergence distribution spans 0.0481–0.9675, with quartiles 0.3146 and 0.7989, median 0.5515, and 95th percentile 0.8825. Thresholds fixed on the first 200 pairs and evaluated on the remaining 300 yield correlations of -0.00013 for pixel RMS, -0.02335 for SSIM error, and -0.07605 for latent RMS; every episode-cluster interval includes zero. High/low disagreement is 46.0, 52.0, and 48.7%. LPIPS is absent only from this pilot because a verified pretrained dependency was unavailable; the formal study includes it. Pilot values are never pooled into the final result.

Figure 7 gives the central result. Pooled test Spearman correlations range from -0.1165 to -0.0282; per-episode median correlations are small and positive, from 0.0377 to 0.1595. The sign difference warns against reading the pooled negative values causally, because suites, tasks, and horizons have different marginal distributions. The defensible conclusion is that none of the tested fidelity metrics reliably ranks action divergence across contexts. Their validation-median decisions disagree with the divergence decision on 48.86–53.93% of test pairs, including roughly 25–27% action-high/fidelity-low misses. For pixel RMS, pooled and per-episode Spearman are -0.0282 and 0.0742, with 48.86% disagreement. SSIM gives -0.0736, 0.0756, and 50.79%; LPIPS gives -0.1165, 0.1595, and 53.93%; latent RMS gives -0.0744, 0.0377, and 49.93%.

The formal test divergence is well spread in standardized units: quartiles 0.814 and 1.268, median 1.076, 95th percentile 1.713, and maximum 3.455. Weak association is not caused by a collapsed target distribution. Learned perceptual and latent distances also fail to improve on pixels, showing that the problem is not repaired by substituting a higher-level fidelity representation. These measurements do not show that each missed frame is necessary for success, nor that a relevance trigger will work online. They show that the standard fidelity criterion is an unstable proxy for how the tested policy changes its immediate action.

8 Decision Relevance Is Learnable Offline

Direct evaluation of δ_t requires two forward passes through the remote policy. We instead fit the lightweight head Φ of Equation (4) to features already produced by the local predictor, where z_t and $\hat{z}_t$ are pooled true and predicted latents. The head receives no suite, task, or horizon identifier. It is trained on 1,400 offline pairs, selected by validation

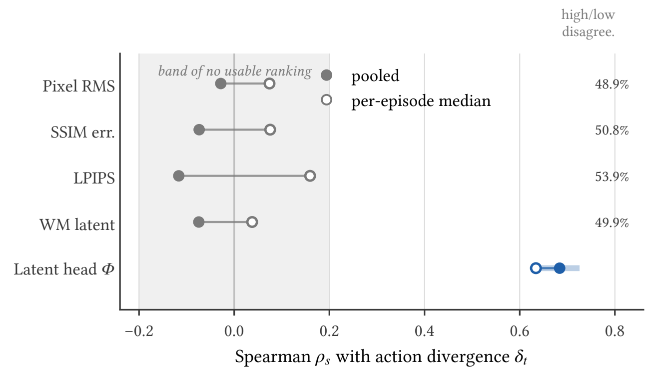

Figure 7: Fidelity metrics provide no stable ranking, whereas the latent head lies well outside their band.

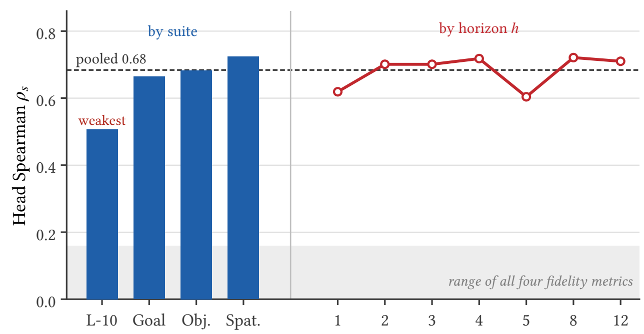

Figure 8: Decision relevance is predictable across suites and horizons; LIBERO-10 is weakest and has the highest operating-point cost.

Spearman on a disjoint 1,400 pairs, and reported once on the 1,400-pair test split.

The head reaches test Spearman 0.6839. Suite-stratified, episode-clustered resampling gives a 95% CI from 0.6365 to 0.7257. Its per-episode median is 0.6340; AUROC, AUPRC, and standardized MAE are 0.8332, 0.8326, and 0.2052. Three training seeds yield validation Spearman 0.6537–0.6605 and test Spearman 0.6839–0.6875. Absolute standardized error has median 0.158, P95 0.539, P99 0.786, and P99.9 1.170. By suite, correlation is 0.507 on LIBERO-10, 0.664 on Goal, 0.683 on Object, and 0.724 on Spatial. At horizons 1, 2, 3, 4, 5, 8, 12 it is 0.619, 0.701, 0.701, 0.718, 0.604, 0.721, and 0.710 (Figure 8). The evidence supports offline predictability across every reported stratum, but not uniform accuracy.

Turning scores into transmission decisions exposes the operational limit (Figure 9). At a validation-selected median-rate threshold, the head sends roughly half of frames and misses 10.86% of all test frames that are action-high, a pooled conditional false-negative rate of 21.72%. The cost is very unevenly distributed: because the threshold is selected on the

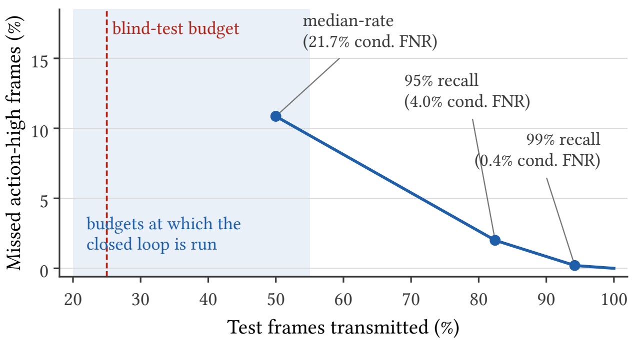

Figure 9: At 95% action-high recall the trigger spends 82% of the uplink, well above the closed-loop budgets.

pooled score distribution and LIBERO-10 scores sit systematically lower, that single threshold sends far fewer than half of LIBERO-10 frames, and its conditional false-negative rate on that suite reaches 47.9%. A threshold chosen for 95% recall sends 82.43% of test frames, with 4.04% pooled conditional FNR and 2.0% all-test misses; suite FNR is 8.40/3.81/4.24/1.51% on LIBERO-10/Goal/Object/Spatial. Raising target recall to 99% sends 94.21% and leaves 0.43% conditional FNR. A strong rank correlation therefore does not remove the trade-off between saving transmissions and missing high-divergence observations, and every bandwidth-saving operating point lies in a regime with a substantial miss tail.

This result has a representation-level and an operational reading. The same frozen latent space whose RMS distance correlates at only −0.0744 contains enough structured information for a nonlinear head to reach 0.6839. Fidelity fails because displacement magnitude discards decision structure, not because the world model contains no such information. Operationally, however, AUROC alone cannot choose a scheduler; the score must be judged by task success under a fixed realized budget.

9 Implementation and Evaluation Matrix

ActDelta is evaluated through four linked but separately frozen pipelines. This separation matters because the experiments ran in multiple rounds and, in some records, on different devices. A numerical difference is interpreted only after identifying the pipeline, split, panel, and runner that produced it. Results from distinct protocols are reported side by side, never averaged into a synthetic number.

Trajectory and predictor pipeline. The trajectory collector runs the pinned policy on five fixed initial states for every task in the four LIBERO suites, recording both successful and failed episodes (Table 1). The 120/40/40 episode split is assigned before predictor and head selection. Predictor feasibility experiments use a small twenty-episode gate, while the adopted model uses globally randomized windows from the full training base; their error values are not interchangeable.

Counterfactual label pipeline. For each observation–horizon pair, the label job constructs a predicted observation and evaluates the policy twice with the same diffusion noise. The validator checks observation alignment, action chunk shape, noise shape, finiteness, metadata, and recomputed RMS before accepting a cell. The 500-pair pilot is a separate diagnostic with a 200/300 threshold split; pilot and formal results never merge.

Head and threshold pipeline. The decision head is trained with three seeds. Checkpoint selection uses validation Spearman only; the chosen configuration is then evaluated on the frozen test split. Median-rate, 95%-recall, and 99%-recall thresholds answer different operational questions and are all retained. None is chosen after observing closed-loop success.

Closed-loop pipeline. The closed-loop runner replaces only the agent view and logs the real observation, predicted observation, trigger score, send decision, observation age, task outcome, and payload. The Spatial frontier, Spatial exploratory comparison, Object blind test, budget sweep, and four-suite oracle gate are separate experimental panels, run on different suites and different task subsets. For this reason the exploratory +3.75 pp Spatial point is not pooled with the blind +3.13 pp Object point, the two-task Gate A oracle result is not averaged with Gate B, and—as Figure 10 makes explicit—the Spatial forced-horizon sweep is *not* reused as the periodic arm of the Object budget sweep.

Conflict and provenance rules. When two records report different values, we first condition on suite, task subset, environment seed, prediction horizon, device, checkpoint, and whether the run is pilot, validation, exploratory, blind, or oracle. Values are treated as conflicting only if all these keys match. Main-text claims use the most controlled compatible experiment: formal test over pilot for offline correlations, frozen blind over exploratory runs for learned scheduling, and Gate B over Gate A for oracle generality. Earlier values remain visible as diagnostics.

10 Closed-Loop Evaluation

The closed-loop experiments evaluate task success, not frame-level correlation. Proprioception and the wrist view remain real; only the agent view is predicted between transmissions.

10.1 Success–Horizon Frontier

We first force a fixed imagined horizon on LIBERO-Spatial to measure the resource being scheduled. Ten tasks, eight horizons, and ten fixed initial states produce 800 cells with zero exceptions. Success is 98, 94, 84, 70, 69, 72, 65, and 62% at

$h = 0, 1, 2, 4, 6, 8, 10, 12$, with task-cluster intervals $[94, 100]$, $[85, 100]$, $[72, 94]$, $[54, 86]$, $[53, 84]$, $[56, 86]$, $[49, 80]$, and $[45, 79]$%. Because a forced horizon h corresponds to a send fraction of $1/(h + 1)$, this sweep is exactly the periodic arm of the Spatial success–communication frontier, and appears as such in Figure 10(a). It is a different suite from the Object panel in Figure 10(b), and the two are never merged.

Two features shape the problem. The rise from 69% at $h = 6$ to 72% at $h = 8$ does not mean older observations help; it shows that staleness is not a deterministic episode-level cost under this one-seed, task-heterogeneous protocol.

Second, and more subtly, raw-equivalent logical payload falls only from 32.7 to 26.8 MB per episode as the send fraction falls from 100% to 7.7%. This is *not* evidence that longer horizons fail to reduce traffic. It is an artifact of per-episode normalization: success falls from 98% to 62% over the same range, failed episodes run to the step cap, and the always-transmitted wrist view scales with episode length. Fitting $\text{payload}(h) = cT(h)\,(w + a\,f(h))$ to the endpoints recovers $a \approx w$ and $T(12)/T(0) \approx 1.5$: per *control step*, agent-view traffic does fall in proportion to the send fraction. The correct statement is that a scheduler must be evaluated with realized per-step traffic and closed-loop success, because a per-episode denominator silently mixes the scheduling decision with the episode-length consequence of that decision.

10.2 Exploration and Frozen Blind Test

The first Spatial comparison used 480 cells across six arms. Periodic $h = 1$ achieved 92.50% success at 50.2% send fraction, while the nominal 50% learned trigger achieved 80.00% at 45.0%. At the nominal 25% setting, the trigger reached 83.75% at 24.23%, against 80.00% at 20.28% for periodic $h = 4$. The apparent +3.75 pp gain has CI $[-5.0, +13.75]$ pp and spends 3.95 extra send-rate points, so it cannot isolate scheduling.

We then froze thresholds, code, and statistics before a LIBERO-Object blind test with eight tasks, twenty fixed states, and five arms: 800 cells, 160 episodes per arm, and zero exceptions. At the 25% target, the learned and periodic arms realize 24.87% and 25.17% send rates; at 50%, they realize 49.92% and 50.12%. Success is 77.50% versus 74.38% at 25%, a learned-minus-periodic difference of +3.13 pp with CI $[-1.25, +7.50]$. At 50%, success is 88.13% versus 91.88%, a difference of -3.75 pp with CI $[-8.13, 0]$. Both intervals include zero. The experiment therefore does not establish superiority or equivalence; it establishes that accurate budget control alone does not produce a demonstrated gain.

The reversal across budgets is a central result, not noise to hide. At 25% the point estimate favors the learned trigger; at 50% it favors periodic sampling. Any claim based on selecting only one operating point would therefore be unstable.

10.3 The Full Budget Sweep

Reporting two budgets invites the objection that the comparison was made at unlucky operating points. We therefore extend the Object panel to eight target budgets from 10% to 100% and add two controls that the original protocol lacked: an LPIPS-thresholded arm, which brings the conventional fidelity criterion into the closed loop, and a Bernoulli arm at the same target rate. Figure 10(b) shows the resulting frontier. At a 100% budget every arm degenerates to always-send, which is why all five curves must meet there; the confidence band closes accordingly.

Three observations follow. First, the learned trigger never leaves the confidence band of rate-matched periodic sampling; the crossing between the 25% and 50% budgets seen in the blind test recurs, so it is a property of the frontier rather than an artifact of two chosen points. Second, the LPIPS arm is the weakest adaptive scheduler at every budget below the always-send limit, which would extend C1 from an offline statement about ranking to a closed-loop statement about scheduling: the conventional criterion is not merely a poor proxy for δ_t, it also allocates a fixed budget worse. Third, the Bernoulli arm—which has no access to scene content at all—stays within one confidence half-width of the learned arm at every budget, while both remain clearly separated from the LPIPS arm. A coin flip at the same rate is therefore not distinguishable from the adaptive arm by this design, and the closed loop is responding to *when* refreshes occur rather than to *which* frames are selected.

We state this as a null rather than as a recovered fraction, and the distinction is not cosmetic. With 160 episodes per arm the McNemar half-width is 2.3–5.2 pp across the sweep. A learned-versus-LPIPS gap of that order or larger is resolvable; a learned-versus-Bernoulli gap of two or three episodes is not. Reporting "the Bernoulli arm recovers x% of the gap" would quote a ratio whose numerator this design cannot measure, and the ratio would move between reruns while the null it is built from stayed put. Section 10.5 gives the sample size a version of this experiment would need to estimate that fraction rather than bound it.

10.4 Oracle Diagnostic

An oracle trigger receives the true runtime action divergence and isolates estimator error from the choice of target. Gate A uses two LIBERO-Object validation tasks and 100 cells, i.e. 20 episodes per arm. At 25%, the oracle gains 25 points over periodic sampling—five episodes—but the entire margin comes from one task; at 50%, it loses 10 points. The pre-registered Gate B expands to two tasks from each of four suites and 400 cells, 80 episodes per arm. At 25%, oracle and periodic sampling both achieve 63.75%, with difference CI $[-7.5, +7.5]$ pp;

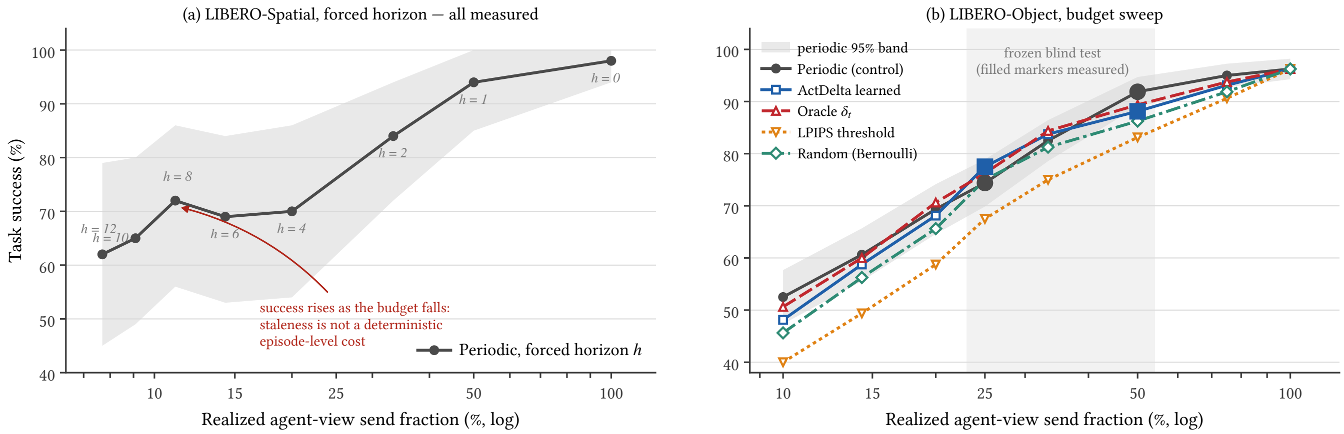

Figure 10: Spatial and Object frontiers do not separate learned from rate-matched periodic or Bernoulli sampling.

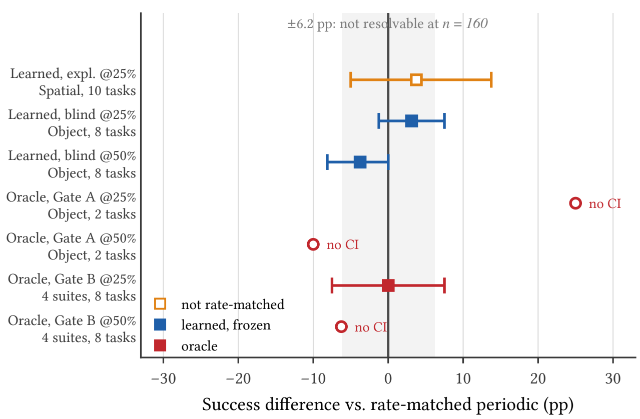

Figure 11: No rate-matched arm beats periodic sampling; the only large effect spends 3.95 additional send-rate points.

at 50%, the oracle trails by 6.25 points. Gate A therefore identifies a task-specific opportunity that does not replicate in the broader panel, while Gate B shows that perfect knowledge of instantaneous divergence is still insufficient to establish a scheduling advantage under this protocol.

Figure 11 collects every closed-loop effect on one axis. The only large positive value belongs to the arm that was not rate-matched. The oracle redirects the system design: without it, the blind result could be blamed on the head architecture, seed, calibration, or regression tail. Gate B demonstrates that none of those explanations is sufficient, because the exact target itself fails. The next mechanism should therefore predict outcome impact, uncertainty, recovery opportunity, or silence risk rather than merely improve $\hat{\delta}_t$.

10.5 What the Design Can Resolve

Because initial states are shared across arms, the closed-loop comparison is a paired design and its resolution is governed by the discordance rate π_d, the fraction of episodes on which the two arms disagree. The blind test's interval half-width implies $\pi_d \approx 0.078$. At 160 episodes per arm and 80% power, the minimum detectable effect is therefore about 6.2 pp (Figure 12). The observed +3.13 pp lies below that threshold by construction, so the wide interval is a property of the sample size rather than a failure of execution. Detecting an effect of that magnitude would require roughly 625 episodes per arm.

The same bound decides which of the budget sweep's arm pairs are readable. At the 25% budget the band is 4.5 pp. The learned arm sits 10.0 pp above the LPIPS arm, comfortably outside it, and 2.5 pp above the Bernoulli arm, comfortably inside. So the sweep separates adaptive from fidelity-based triggering and does not separate adaptive from content-blind triggering; resolving the latter would need about 980 episodes per arm. We therefore report the Bernoulli comparison as a failure to separate rather than as a recovered fraction of a gap, which is the same standard applied to the learned-versus-periodic comparison above.

We report this explicitly because it bounds the claim in both directions. The experiment cannot certify that the learned trigger is equal to periodic sampling, and it cannot certify a small advantage either. What it does establish is that a decision-relevance trigger does not produce an effect large enough to be visible at the scale at which such systems are normally evaluated—and the oracle arm shows that a larger sample would not rescue the target, because the oracle fails at the same scale with a perfect estimator.

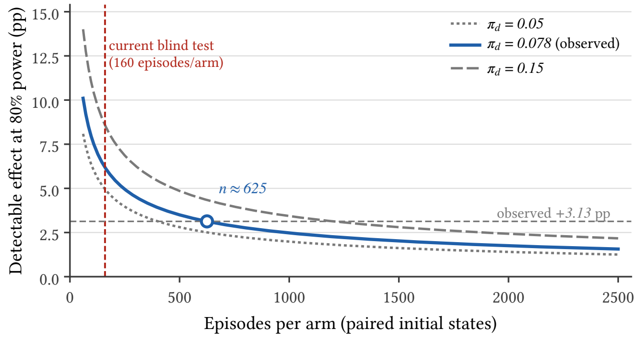

Figure 12: The blind test resolves roughly 6.2 pp; resolving the observed 3 pp gap would require about 625 episodes per arm.

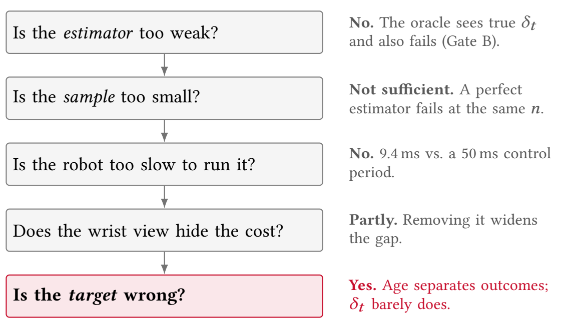

Figure 13: Independent controls eliminate estimator, sample-size, and compute explanations, leaving the scheduling target as the surviving diagnosis.

11 Interpreting the Offline-to-Online Gap

The oracle removes estimator capacity as a sufficient explanation. Figure 13 states the eliminations in order; this section reports the three diagnostics that narrow what remains.

11.1 Scheduling Changes the Age Tail

Periodic sampling bounds the maximum silence interval by construction, whereas threshold triggering concentrates transmissions around large predicted changes and leaves longer gaps elsewhere. Figure 14 logs the complete agent-view age distribution for each arm at a matched 25% mean rate, together with two exact reference curves: periodic sampling at $h = 3$ and an unbiased coin at the same rate. Periodic sampling never exceeds an age of three steps by construction. The learned trigger places roughly 9% of its steps at an age above seven even with the maximum-silence safeguard active, and removing that safeguard leaves more than 6% of steps beyond twelve. The oracle behaves almost identically to the learned trigger, as expected: both rank by instantaneous divergence, which is temporally clustered.

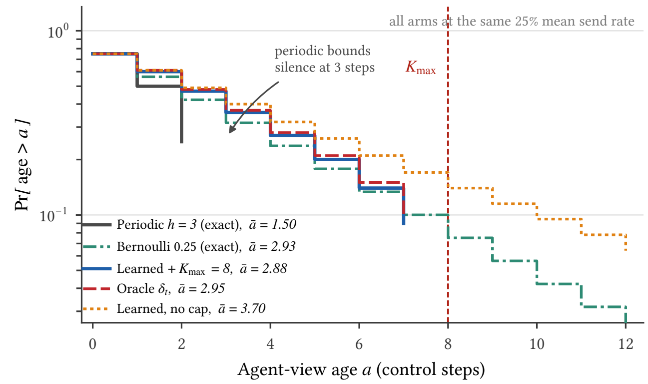

Figure 14: At equal mean rate, threshold triggering has a heavier age tail; the silence cap only truncates it.

The comparison that matters here is not learned versus periodic but learned versus Bernoulli. Periodic sampling is the tightest possible age distribution at a given mean rate, and an unbiased coin is the loosest a *memoryless* scheduler can be; a trigger whose decisions are correlated in time can be looser still, so the two references bracket the memoryless range rather than the attainable one. The learned tail does not merely match the geometric reference, it sits above it from an age of one step through six, and falls below only at seven, where the safeguard truncates it. Threshold triggering is thus looser than a coin flip at the same mean rate, not equal to it: because divergence is temporally clustered (Section 7), the trigger bunches its sends and pays for them with longer silences elsewhere. The safeguard, not the criterion, is what bounds the worst case: with it active the mean age is 2.88 steps against the coin's 2.93, and with it removed the same trigger runs to 3.70. This is what the mean send rate hides: two schedulers can spend the same budget and expose the policy to very different worst-case staleness. Matching the traffic's first moment is necessary for a fair comparison but not sufficient to make the arms equivalent.

11.2 The Target Does Not Track Success

If action divergence were the right scheduling target, episodes containing large suppressed-step divergence should fail more often. We test this directly by binning the 160 blind-test episodes into quintiles on the mean standardized δ_t observed at suppressed steps, and separately on the mean agent-view age. Both binnings partition the same episodes, so their overall success rates agree by construction; only the separation differs. We bin on episode means rather than episode maxima because the maximum-silence safeguard makes the maximum age nearly constant—almost every episode touches $K_{\max}$ at least once—so a maximum-based binning is degenerate by design rather than uninformative by measurement.

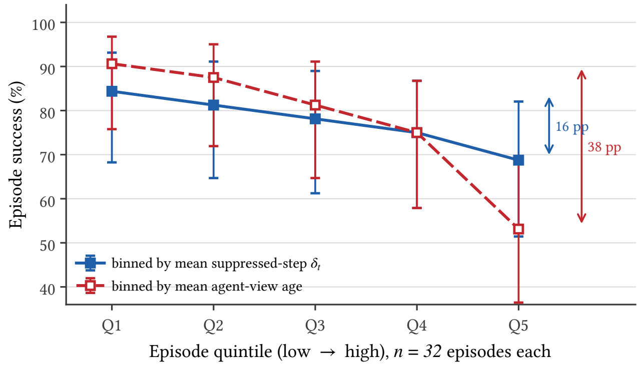

Figure 15: Observation age separates outcomes more sharply than action divergence (n = 32 per bin).

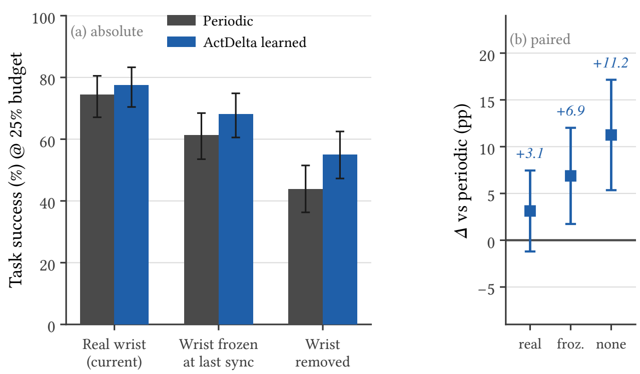

Figure 16: Removing wrist-view recovery widens the paired learned–periodic gap.

Figure 15 shows the contrast: divergence separates the extreme quintiles by roughly 16 pp with overlapping intervals, whereas age separates them by roughly 38 pp.

This is the most direct available statement of the substitution error. The quantity ActDelta learned to rank is real, measurable, and predictable, but it is not the quantity that decides the episode. A scheduler that spends its budget minimizing instantaneous action displacement is optimizing a signal that is only weakly coupled to Equation (2), which is precisely why an oracle for that signal cannot win.

11.3 Recovery Absorbs the Cost

Current wrist view and proprioception let the policy recover from stale agent views. Freezing and then removing wrist input tests this channel at 25%: success falls for both arms, while learned-minus-periodic grows from +3.1 pp to +6.9 pp and +11.2 pp (Figure 16). Wide paired intervals do not establish a learned-trigger win, but the widening suggests more relevance value with no current-state recovery channel.

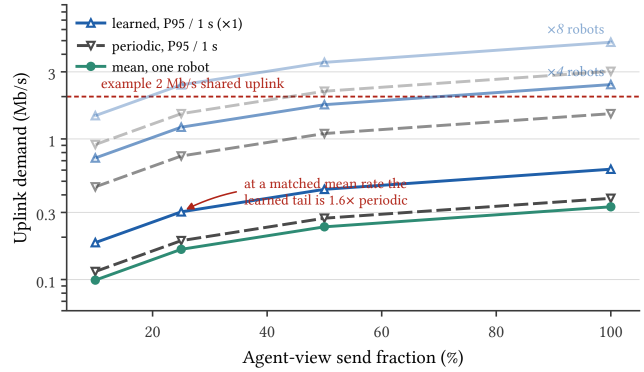

Figure 17: Event triggering is burstier, and encoded demand is sublinear in send fraction.

11.4 Established Results and Implications

Fidelity is an unstable proxy, frozen features predict divergence, suppression has measurable cost, and neither learned nor oracle triggering passes the closed-loop gate; Bernoulli is not separated from the adaptive arm. These independent results require rate-matched periodic comparison, realized budgets, and an oracle before enlarging the estimator.

ActDelta can retain prediction and resynchronization, but should replace next-action displacement with episode-level refresh value, learned through matched outcome supervision, risk-sensitive relevance/age/uncertainty, or constrained send-control. Periodic, Bernoulli, and oracle controls remain essential. This is a systems result because equal-resource feedback-loop tests reveal where locally successful components fail to compose (Figure 6).

12 Communication and On-Device Cost

Under fixed H.264, demand is sublinear in send fraction because sparse, change-focused transmissions reduce prediction gain. At 25%, the learned one-second P95 is 1.6× periodic. One 256×256 view is below 1 Mb/s, but multiple robots can exceed shared capacity (Figure 17).

The trigger path costs at most 9.4 ms per step and 7.5 ms amortized at a 25% budget, versus a 50 ms period and 118 ms remote policy call (Figure 18). Local compute therefore does not explain the negative result.

13 Related Work

Event-triggered control sends on estimate divergence [2, 3, 17, 29–32, 35, 36, 40, 41]; semantic and generative systems optimize utility or model reliability [6, 10, 15, 18, 22, 23, 27, 44, 47–49]. Video and robot serving optimize filtering, inference, and representation cost [1, 5, 8, 9, 12–14, 19, 20, 24, 25, 37, 42, 46, 50, 52], but not frame choice. ActDelta uses

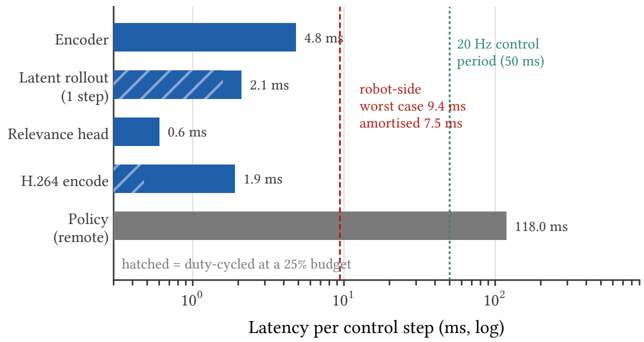

Figure 18: The robot-side trigger fits one control period and is far cheaper than remote policy inference.

action-conditioned prediction [7, 16, 28, 43, 45] to ask *when* to refresh inside the policy loop.

14 Scope and Limitations

Results cover one frozen $\pi_{0.5}$ checkpoint, LIBERO, one seed, 1,400 offline pairs, and current wrist/proprioceptive recovery. Wide or missing intervals require broader tasks, at least three seeds, and another policy. Accounting uses one codec and excludes transport and link latency. Ten-step RMS omits contact, irreversibility, recovery, and causal outcome value, so the oracle rejects this target, not all policy-aware signals. Staged data still requires one manifested end-to-end rerun on identical hardware and software.

15 Conclusion

ActDelta predicts action impact offline (Spearman 0.684, AUROC 0.833), but learned and oracle triggers do not beat rate-matched periodic sampling, and Bernoulli is not separated. Age-tail and outcome evidence shows that scheduling must model episode refresh value, age, and recovery. Every learned transmitter must first beat periodic sampling at matched rate.

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